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Aryabhatta Knowledge University, Patna

B.Tech 4th Semester Exam., 2019

Time: 3 hours

Code: 031403 ***Subject: ELECTRICAL MACHINES-II*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer (any seven): **[2x7=14]**

(a) The pitch factor for a two-thirds short-pitch coil is
(i) 0.5 (iii) 0.866
(ii) 0.66 (iv) 0.707

(b) The speed regulation of a synchronous motor is
(i) unity (iii) infinity
(ii) zero (iv) always less than one

(c) The armature flux helps the main field flux when the load power factor is
(i) unity (iii) 0.8 lagging
(ii) zero-lagging (iv) zero-leading

(d) In alternators, damper windings are used to
(i) reduce eddy-current loss (iii) make the rotor dynamically balanced
(ii) prevent hunting (iv) reduce armature reaction

(e) In a split-phase motor, the running winding should have
(i) high resistance and low inductance (iii) low resistance and high inductance
(ii) high resistance and high inductance (iv) low resistance and low inductance

(f) In a capacitor start and run motors, the function of the running capacitor in series with the auxiliary winding is to
(i) improve power factor (iii) increase overload capacity
(ii) reduce fluctuations in torque (iv) improve torque

(g) In case the field of a synchronous motor is under excited, the power factor will be
(i) leading (iii) zero
(ii) lagging (iv) unity

(h) When the excitation of an unloaded salient pole synchronous motor gets disconnected
(i) the motor will burn
(ii) the motor will stop
(iii) the motor will run as a reluctance motor at the same speed
(iv) the motor will run as a reluctance motor at a lower speed

(i) The back e.m.f. setup in the stator of a synchronous motor will depend on
(i) rotor speed only (iii) rotor excitation and speed
(ii) rotor excitation only (iv) coupling angle, rotor speed and excitation

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(j) If two mechanically coupled alternators deliver power at 50 Hz and 60 Hz respectively, then the highest speed of alternators will be

(i) 1200 r.p.m. (iii) 600 r.p.m.

(ii) 1500 r.p.m. (iv) 300 r.p.m.

Q.2 Solve both questions : **[7+7=14]**

(a) Explain the armature reaction phenomenon in a cylindrical rotor synchronous generator by drawing the space and time phasor diagram when operating at an internal power factor of (i) zero-lagging and (ii) zero-leading.

(b) For a salient pole, machine P_1 and P_2 are the maximum values of electromagnetic power and reluctance power respectively. Show that the load angle δ, at which the resultant power is maximum can be obtained from the relation $\cos\delta=[-P_1\pm\sqrt{(P_1^2+ 32\ P_2^2)}] / [8P_2]$

Q.3 Solve both questions : **[7+7=14]**

(a) A 6-pole, 3-phase, 50 Hz alternator has 12 slots per pole and 4 conductors per slot. The winding is five-sixth pitch and the flux per pole is 1.5 Wb. The armature coils are all connected in series with star connection. Calculate the induced e.m.f. per phase.

(b) Define the various modes of excitation of synchronous generator. Also draw the V curve and inverted V curve for such system.

Q.4 Solve both questions : **[7+7=14]**

(a) Discuss Blondel's two-reaction theory of salient pole synchronous machine.

(b) The speed regulations of two 800 kW alternators A and B running in parallel are 100% to 104% and 100% to 105% from full-load to no-load respectively. How will the two alternators share a load of 1000 kW?

Q.5 Solve both questions : **[6+8=14]**

(a) State the need for parallel operation of alternators. What are the conditions for parallel operation of three-phase alternators?

(b) A 500-kVA, 1.1-kV, 50 Hz, three-phase star-connected alternator has an effective armature resistance of 0.2 ohm per phase. The synchronous reactance per phase is 1.5 ohm. Find the full-load voltage regulation at (i) 0.8 lagging power factor and (ii) 0.8 leading power factor. Draw phasor diagram in each case.

Q.6 Solve both questions : **[6+8=14]**

(a) Explain why a synchronous motor does not have starting torque. Explain one method of starting a synchronous motor.

(b) An industrial plant has a load of 800 kW at a power factor of 0.8 lagging. It is desired to install a synchronous motor to deliver a load of 200 kW and also serve as a synchronous condenser to improve the overall power factor of the plant to 0.92. Determine the kVA rating of the synchronous motor and its power factor. Assume that the motor has an efficiency of 90 percent.

Q.7 Solve both questions : **[7+7=14]**

(a) Explain how the stationary, pulsating m.m.f. wave of a single-phase induction motor can be considered as equivalent to two equal but oppositely rotating m.m.f. waves. Also explain why the forward flux wave is several times greater than the backward flux wave at normal rotor speed.

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(b) A 230-V, 4-pole, 50 Hz capacitor start a single-phase induction motor has the following constants and losses: r_1= 2.10Ω, r_2= 4Ω, X_m= 70Ω, x_1= 3Ω, x_2= 3Ω. Core loss = 48 watts, Friction and windage loss = 20 watts. For a slip of 0.05, compute the stator current, pf, power output, torque and efficiency when this motor is running at rated voltage and frequency with its auxiliary winding open.

Q.8 Solve both questions : **[6+8=14]**

(a) Draw the complete torque-speed characteristics of a single-phase induction motor when no auxiliary winding is provided on the stator. Show how the torque speed characteristics is modified when an auxiliary winding is provided.

(b) Mention the conditions necessary for the production of rotating magnet field with the help of stationary windings.

Q.9 Solve both questions : **[7+7=14]**

(a) What is a universal motor? How is it different from a d.c. series motor? Mention its applications.

(b) Explain the construction and working principles of a repulsion motor. How is a repulsion-induction motor different from a repulsion motor?

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